

Notes: Lou (RFS, 2014)
Attracting Investor Attention through Advertising

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1 Big picture

- **Question.** Can product-market advertising attract investor attention, temporarily push up stock prices, and be strategically adjusted by managers before equity sales?
- **Core hypothesis.** Advertising is intended to make consumers aware of products, but it can also make investors aware of the firm's stock. Because retail investors are attention constrained and buy attention-grabbing stocks more readily than they sell them, advertising can create temporary buying pressure.
- **First main result.** Firms with large increases in advertising spending experience high contemporaneous abnormal returns and then low subsequent abnormal returns. The paper interprets this inverted-V return pattern as temporary overpricing from investor attention.
- **Second main result.** Retail investors are net buyers of stocks with higher advertising growth, while short interest rises at the same time, suggesting that more sophisticated investors partly recognize the transitory overvaluation.
- **Third main result.** Advertising increases before and during top-level insider selling and falls afterward. Similar patterns occur around seasoned equity offerings and stock-financed acquisitions, but not around debt issues or cash-financed acquisitions.
- **Mechanism.** Advertising before insider sales is less effective in generating future sales, suggesting that advertising in these periods is less product-market motivated and more investor-attention motivated.
- **Economic takeaway.** Advertising is not only a real investment in product demand or brand capital. It can also be a financial-market attention technology that managers may use opportunistically when short-term stock prices matter.

2 Incremental contribution of the paper

- **Product-market action as financial-market attention shock.** The paper extends attention-based asset-pricing research by showing that a real product-market decision, advertising spending, moves investor attention and stock returns.
- **Return pattern rather than ownership/liquidity only.** Prior advertising-finance papers emphasize investor base and liquidity. This paper documents a full return cycle: contemporaneous overperformance followed by reversal.
- **Direct retail-trading evidence.** The paper connects advertising growth to small-trade order imbalance, making the attention mechanism more direct than simply observing returns.
- **Managerial opportunism channel.** The paper links the temporary return effect to managerial choices around insider sales, equity issuance, and stock-financed acquisitions, connecting investor attention to corporate policy.
- **Agency cost of advertising.** The future-sales test shows that advertising around insider sales is less productive in generating sales, so attention-motivated advertising can be a real investment distortion.
- **Instrumental-variable evidence.** Restricted share vesting provides a plausibly predetermined shock to insider selling, helping separate advertising-triggered sales from sales-timed advertising campaigns.
- **Placebo structure.** Lower-level insiders, debt issues, and cash acquisitions serve as placebo-style tests because they should not generate the same incentive to boost the stock price through advertising.

3 Data and measurement

3.1 Advertising and firm characteristics

- Advertising expenditures come from Compustat annual item `data45`; total assets, sales, and capital expenditures come from Compustat items `data6`, `data12`, and `data128`.
- The sample spans 1974–2010, beginning when Compustat advertising data become available.
- Firm-year observations with missing advertising spending are excluded in the baseline, although robustness treats missing advertising as zero.
- The Compustat sample is merged with CRSP monthly stock returns, market capitalization, and trading volume.
- Change variables and ratio variables are winsorized at the 1st and 99th percentiles to reduce outlier influence.

Interpretation: Table 1 shows that firms reporting advertising are slightly larger than firms with missing advertising data, but the two groups have similar age and market-to-book ratios. Advertising is economically meaningful: mean annual advertising is about \$42 million, 4.33% of sales, and 4.73% of assets.

3.2 Investor attention and trading measures

- Retail trading is proxied with small-order imbalances from TAQ and ISSM data.
- A small order is below \$5,000, and monthly imbalance is small buy orders minus small sell orders divided by total small orders.
- The small-trade classification is mainly used pre-2000 because decimalization later caused institutions to split trades more often.
- Short interest is measured as monthly change in short interest relative to shares outstanding.
- Institutional ownership comes from Thomson Financial CDA/Spectrum and is total institutional shares divided by shares outstanding.

Interpretation: The attention mechanism predicts that advertising growth attracts net retail buying. The short-interest test asks whether more sophisticated investors arbitrage against the temporary price pressure.

3.3 Insider sales, vesting, and corporate events

- Insider trading data come from Thomson Financial Insider Filing, using SEC Forms 3, 4, and 5.
- Top-level insiders include the chairman, chief executive officer, chief operating officer, general counsel, and president.
- Lower-level insiders include vice chairmen, committee members, CFOs, treasurers, secretaries, CTOs, and related roles.
- Insider sales are aggregate shares sold by top executives in a year divided by shares outstanding.
- Restricted-share vesting data begin in 2006 and are used as an instrument for insider selling.
- Seasoned equity offerings, debt issues, stock-financed acquisitions, and cash-financed acquisitions are from SDC and scaled by market capitalization.

Interpretation: The event data allow the paper to test whether advertising rises when stock prices matter most to managers: before insider sales, equity issuance, and stock-financed acquisitions.

4 Empirical specifications

4.1 Portfolio sort on advertising growth

Each month, firms are sorted into deciles based on changes in log advertising spending in the previous fiscal year. Portfolios are rebalanced monthly when new advertising data become available and are held for two years.

$$\Delta AD_{i,t} = \log(AD_{i,t}) - \log(AD_{i,t-1}).$$

Interpretation: The portfolio test is designed to detect the inverted-V pattern predicted by temporary attention-induced overpricing: high returns in the formation year followed by reversals.

4.2 Fama-MacBeth return regressions

$$R_{i,s} = \alpha_s + \beta_s \Delta AD_{i,t} + \Gamma'_s \text{Controls}_{i,t} + \varepsilon_{i,s},$$

where $R_{i,s}$ is either monthly stock return or earnings-announcement return in year 1 or year 2 after fiscal-year end.

Controls include asset growth, sales growth, capex growth, cumulative stock returns, firm size, book-to-market, firm age, turnover, and other return predictors.

Interpretation: These regressions test whether advertising-growth effects survive known return predictors and whether they show up around earnings announcements.

4.3 Retail order imbalance and short interest

$$IMBAL_{i,s} = \alpha + \beta \Delta AD_{i,t} + \Gamma'_s \text{Controls}_{i,t} + \varepsilon_{i,s}.$$

$$\Delta \text{ShortInterest}_{i,s} = \alpha + \beta \Delta AD_{i,t} + \Gamma'_s \text{Controls}_{i,t} + \varepsilon_{i,s}.$$

Interpretation: The retail-imbalance coefficient tests the investor-attention channel directly. The short-interest coefficient asks whether arbitrageurs respond to inflated valuations.

4.4 Cross-sectional mechanism regressions

$$R_{i,s} = \alpha + \beta \Delta AD_{i,t} + \theta D_i + \phi(\Delta AD_{i,t} \times D_i) + \Gamma'_s \text{Controls}_{i,t} + \varepsilon_{i,s}.$$

The dummy D_i captures consumer-product firms, high analyst coverage, high retail buying, high turnover, high institutional ownership, high brand count, or Craigslist-advertising similarity.

Interpretation: If the mechanism is investor attention, return reversals should be stronger where advertising is more visible to retail investors and where other information channels are weaker.

4.5 Advertising around insider sales

$$AD_{i,t} = \alpha_i + \lambda_t + \beta_1 \text{PreEvent}_{i,t} + \beta_2 \text{Event}_{i,t} + \beta_3 \text{PostEvent}_{i,t} + \Gamma'_s \text{Controls}_{i,t} + \varepsilon_{i,t}.$$

Event years are years with top-level insider selling. PreEvent, Event, and PostEvent mark the adjacent years.

Interpretation: The opportunistic advertising hypothesis predicts $\beta_1 > 0$, $\beta_2 > 0$, and $\beta_3 < 0$: a sharp rise before or during insider sales and a drop afterward.

4.6 Propensity matching and instrumenting insider sales

The matching design compares event firms to non-event firms matched on industry, size, book-to-market, and past returns. The IV design uses restricted-share vesting as an instrument for insider sales:

$$Event_{i,t} = \pi_0 + \pi_1 Vesting_{i,t} + \Gamma' Controls_{i,t} + u_{i,t}.$$

Interpretation: The matching design addresses functional-form concerns; the vesting instrument addresses reverse timing, because vesting schedules are set before current advertising choices.

4.7 Sales-growth and placebo event tests

$$\Delta Sales_{i,t+1} = \alpha + \beta \Delta AD_{i,t} + \theta Event_{i,t} + \phi(\Delta AD_{i,t} \times Event_{i,t}) + \Gamma' Controls_{i,t} + \varepsilon_{i,t+1}.$$

Interpretation: If advertising near insider sales is opportunistic rather than product-market motivated, the usual positive link between advertising and future sales should weaken in event years.

5 Identification strategy

5.1 Identifying a temporary attention effect

- The return pattern is an inverted V: advertising growth predicts high contemporaneous returns and low future returns.
- The result survives factor adjustments, DGTW characteristic adjustment, subsamples, and industry-adjusted advertising growth.
- Retail order imbalance rises with advertising growth, consistent with retail attention rather than only fundamentals.
- Short interest rises contemporaneously, consistent with sophisticated investors recognizing overpricing.

Interpretation: The attention channel is not identified by one regression alone. It is identified by the combination of return reversal, retail buying, short selling, and cross-sectional heterogeneity.

5.2 Separating attention from investment or signaling stories

- A pure investment story predicts advertising should forecast higher future cash flows and returns, not reversals.
- Table 5 shows the negative future-return relation is stronger in consumer-product firms, low-analyst firms, high-retail-buying firms, high-turnover firms, low-institutional-ownership firms, and firms with fewer distinct product brands.
- These patterns match attention better than investment opportunities.
- Table 8 shows that advertising generally predicts future sales, but not in insider-selling years.

Interpretation: The sales result is especially important because it shows that advertising near insider sales does not function like normal demand-generating advertising.

5.3 Identifying opportunistic managerial behavior

- Insider sales provide periods when managers personally benefit from a temporarily higher stock price.
- Advertising rises before and during top-level insider sales and falls afterward.
- Propensity matching shows the pattern relative to similar firms without insider sales.
- Restricted-share vesting provides a plausibly predetermined source of insider-selling pressure and produces similar advertising responses.
- Lower-level insiders show no significant advertising pattern, consistent with lack of control over advertising decisions.
- Equity issues and stock-financed acquisitions show the same advertising pattern; debt issues and cash acquisitions do not.

Interpretation: The paper does not have randomized advertising shocks. Instead, it builds a chain of evidence consistent with attention and opportunism while ruling out several alternatives.

5.4 Remaining limitations

- Advertising choices are endogenous and annual Compustat advertising data are coarse.
- The paper cannot directly observe managers' private intent.
- Insider sales may still be timed around planned product campaigns, though vesting evidence reduces this concern.
- The small-trade retail imbalance measure is most credible pre-decimalization.

Interpretation: The identification is best understood as cumulative and triangulated rather than based on a single natural experiment.

6 Tables and figures

6.1 Table 1: Summary statistics and Table 2: Calendar-time portfolio returns

Read: Table 1 shows the sample covers 1974–2010 and that average advertising spending is about \$42 million, 4.33% of sales, and 4.73% of assets. Table 2 shows the central inverted-V return pattern: the high-minus-low advertising-growth portfolio earns strong positive returns in the formation year and negative returns in the next two years.

Interpretation: Table 1 establishes that advertising is economically meaningful and that event variables are sizable. Table 2 is the first return evidence: advertising-growth stocks temporarily outperform and then reverse.

Economic message: Product-market advertising can produce financial-market price pressure, not only product-market demand.

Table 1
Summary statistics

Panel A: Advertising spending (1974–2010)

Firm characteristics	Firms with AD spending			Firms missing AD spending		
	Mean	Median	SD	Mean	Median	SD
Market cap (million \$)	1,784	93	10,975	1,170	95	7,986
Total assets (million \$)	3,034	163	33,884	2,828	133	27,028
Sales (million \$)	1,480	120	8,120	1,102	109	5,733
Net earnings (million \$)	76.58	3.73	799.46	57.45	3.33	553.52
Firm age	15.31	12.00	11.85	15.57	12.00	12.29
Market to book	2.48	1.52	3.42	2.65	1.46	3.91
Advertising (million \$)	42.09	2.16	219.14			
AD-to-sales ratio	4.33%	2.06%	7.33%			
AD-to-assets ratio	4.73%	2.27%	7.27%			
% growth in AD	27.56%	10.16%	84.23%			
No obs.		66,113			122,370	

Panel B: Various corporate events

	From	Obs.	All Obs. with AD	Mean	Median	SD
Insider sales and restricted shares vesting						
Insider sales	1986	9,005	45,034	1.22%	0.48%	2.21%
Restricted shares vesting	2006	2,170	9,275	0.20%	0.10%	0.27%

Other corporate events

	1974	3,458	66,113	17.74%	13.50%	17.86%
Equity issues						
Debt issues	1974	4,979	66,113	26.47%	15.74%	35.02%
Stock-financed acquisitions	1980	490	55,689	36.07%	16.41%	45.73%
Cash-financed acquisitions	1980	2,869	55,689	12.53%	7.30%	14.13%

This table reports summary statistics of the sample that spans the period 1974 to 2010. Panel A: This panel is the summary of firms with missing and nonmissing advertising spending data. Data on advertising spending (data45), total assets (data6), equity (data216), annual sales (data12), income (data18), and firm age are obtained from the COMPUSTAT annual files. The COMPUSTAT sample is then merged with CRSP monthly stock files to obtain market capitalization. To reduce the impact of outliers, variables—the market-to-book ratio, advertising-spending-to-sales ratio, advertising-spending-to-assets ratio, and annual growth in advertising spending—are winsorized at the 1st and 99th percentiles. To further ensure data quality, the first two years of financial data of each firm from COMPUSTAT are excluded from the sample. Panel B: This panel contains the summary for various corporate events, including insider selling, restricted shares vesting, equity issues, debt issues, and stock- and cash-financed acquisitions (only acquirers are considered). Insider sales (restricted shares vesting) are defined as the aggregate number of shares sold (vested) across all top executives over a year scaled by the shares outstanding. Equity issues, debt issues, and stock- and cash-financed acquisitions are defined as the total principal amount involved in these transactions over a year divided by the market capitalization of the firm. All

(a) Table 1: Summary statistics.

Table 2
Calendar-time portfolio returns

Decile	Excess	3-factor alpha	4-factor alpha	DGTV adjusted	Excess	3-factor alpha	4-factor alpha	DGTV adjusted	Excess	3-factor alpha	4-factor alpha	DGTV adjusted
Year 0: the formation year												
Panel A: Sort by AAD, 1974–2010												
1	-0.13%	-0.49%	-0.39%	-0.31%	0.79%	-0.07%	0.11%	0.10%	1.18%	0.31%	0.43%	0.22%
10	0.04%	0.72%	0.57%	0.56%	0.20%	-0.56%	-0.17%	-0.16%	0.30%	-0.40%	-0.14%	-0.39%
10 - 1	1.07%	1.21%	0.96%	0.87%	-0.80%	-0.60%	-0.48%	-0.44%	-0.82%	-0.71%	-0.57%	-0.62%
	(0.72)	(0.72)	(0.59)	(0.50)	(-3.54)	(-3.24)	(-3.07)	(-3.13)	(-4.52)	(-3.88)	(-3.38)	(-3.70)
Panel B: Sort by industry-adjusted AAD, 1974–2010												
1	0.15%	-0.39%	-0.36%	-0.36%	0.72%	-0.09%	0.32%	0.01%	1.26%	0.46%	0.55%	0.31%
10	0.09%	0.50%	0.52%	0.41%	0.16%	-0.41%	-0.27%	-0.40%	0.38%	-0.42%	-0.29%	-0.40%
10 - 1	0.04%	0.89%	0.88%	0.77%	-0.69%	-0.72%	-0.59%	-0.41%	-0.88%	-0.88%	-0.64%	-0.71%
	(0.74)	(0.91)	(0.86)	(0.56)	(-3.55)	(-3.15)	(-3.22)	(-2.62)	(-5.57)	(-5.55)	(-4.85)	(-4.66)
Panel C: Sort by AAD, 1974–1993												
1	0.01%	-0.39%	-0.40%	-0.38%	0.65%	-0.03%	0.26%	0.03%	1.00%	0.28%	0.47%	0.14%
10	0.72%	0.50%	0.73%	0.59%	0.05%	-0.66%	-0.42%	-0.43%	0.07%	-0.57%	-0.25%	-0.57%
10 - 1	1.20%	1.29%	1.13%	0.97%	-0.66%	-0.62%	-0.68%	-0.46%	-0.91%	-0.85%	-0.71%	-0.71%
	(0.92)	(1.21)	(0.82)	(0.54)	(-3.04)	(-3.19)	(-3.28)	(-2.70)	(-4.80)	(-4.60)	(-3.86)	(-3.94)
Panel D: Sort by AAD, 1996–2010												
1	-0.16%	-0.47%	-0.38%	-0.32%	0.86%	-0.08%	0.32%	0.06%	1.12%	0.43%	0.43%	0.24%
10	0.72%	0.50%	0.40%	0.30%	0.13%	-0.50%	-0.08%	-0.14%	0.37%	-0.21%	-0.11%	-0.46%
10 - 1	0.89%	0.97%	0.78%	0.70%	-0.65%	-0.43%	-0.40%	-0.41%	-0.77%	-0.64%	-0.54%	-0.64%
	(0.77)	(0.71)	(0.48)	(0.30)	(-2.40)	(-2.25)	(-2.10)	(-2.30)	(-2.40)	(-2.37)	(-2.22)	(-2.30)

This table shows calendar-time equal-weighted monthly returns to portfolios sorted by changes in log advertising spending (AAD). Panel A: Portfolios are ranked by changes in advertising spending in the previous fiscal year and are rebalanced every month using the most recent advertising spending data. The portfolios are then held for two years. Panel B: Portfolios are sorted by industry-adjusted changes in advertising spending from the previous fiscal year. Panels C and D: These two panels report subsample analysis, where portfolios are ranked by changes in advertising spending in the previous fiscal year. Years 1994 and 1995 are excluded from the analysis because of a regulatory change on the reporting of advertising expenditures introduced in 1994. To deal with overlapping portfolios in each holding month, the equal-weighted average returns across portfolios formed in different months is reported (as Jegadeesh and Titman 1993). A number of risk benchmarks are employed here: the Fama-French three-factor model, the Carhart four-factor model, and the Daniel et al. (2006, DGTV) characteristics-adjustment model. *t*-statistics, shown in parentheses, are computed based on standard errors corrected for serial dependence with twelve lags. 5% statistical significance is indicated in bold.

(b) Table 2: Calendar-time portfolio returns.

6.2 Table 3: Fama-MacBeth return regressions and Table 4: Retail investor order imbalance and short interest

Read: Table 3 shows that advertising growth negatively predicts future stock returns and earnings-announcement returns after controls. Table 4 shows advertising growth predicts retail net buying and higher short interest.

Interpretation: Table 3 shows that the reversal is not just a portfolio-sort artifact. Table 4 directly supports the mechanism: retail investors buy attention-grabbing advertising stocks, while short sellers partly bet against the overvaluation.

Economic message: Advertising creates disagreement: retail demand pushes prices up while arbitrageurs recognize temporary overpricing.

Table 3
Fama-MacBeth return regressions

(X 100)	Monthly stock returns in year 1		Monthly stock returns in year 2		Earnings annmnt returns in year 1		Earnings annmnt returns in year 2	
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
ΔAD	-0.24*** [0.07]	-0.11** [0.05]	-0.21*** [0.05]	-0.11*** [0.04]	-0.32** [0.15]	-0.18* [0.10]	-0.35*** [0.14]	-0.20** [0.09]
$\Delta ASSETS$		-0.79*** [0.27]		-1.23*** [0.23]		-0.18 [0.26]		-0.96 [1.02]
$\Delta SALES$		-0.18 [0.31]		0.26 [0.25]		1.08 [0.71]		1.03 [1.08]
$\Delta CAPEX$		-0.08 [0.06]		-0.08 [0.07]		-0.14 [0.16]		-0.51 [0.30]
Other controls	yes	yes	yes	yes	yes	yes	yes	yes
Adj. R^2	0.07	0.09	0.06	0.08	0.12	0.16	0.11	0.16
No obs.	386,488	386,488	351,985	351,985	76,588	76,588	76,627	76,627

This table reports forecasting regressions of stock returns on changes in advertising spending and other control variables that are known to predict stock returns. The dependent variable in Columns 1 and 2 is the monthly stock return in the year following the fiscal year-end; in Columns 3 and 4, it is the monthly stock return in the second year following the fiscal year-end; in Columns 5 and 6, it is the quarterly earnings announcement return in the year following the fiscal year-end; and in Columns 7 and 8, it is the quarterly earnings announcement return in the second year following the fiscal year-end. The independent variable of interest is the change in log advertising spending (ΔAD) in the previous fiscal year. $\Delta ASSETS$, $\Delta SALES$, and $\Delta CAPEX$ are changes in log assets, sales, and capital expenditures in the same fiscal year. Other control variables include cumulative stock returns at various horizons, firm size, the market-to-book ratio, firm age, and aggregate equity issuance in the previous four years as defined by [Daniel and Titman \(2004\)](#), discretionary accruals as defined by [Kis \(2001\)](#), and average monthly turnover in the previous year. Coefficients are estimated using the Fama-MacBeth approach. Standard errors, shown in brackets, are corrected for serial-dependence with twelve lags. *, **, and *** denote statistical significance at the 90%, 95%, and 99% level, respectively.

pattern is unrelated to common risk factors and previously known return

(a) Table 3: Fama-MacBeth return regressions.

Table 4
Retail investor order imbalance and short interest

	Monthly ImbNum in year 0		Monthly ImbNum in year 1		$\Delta Short Interest$ in year 0		$\Delta Short Interest$ in year 1	
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
ΔAD	0.10*** [0.01]	0.04*** [0.01]	0.07*** [0.02]	0.02* [0.01]	0.06*** [0.02]	0.04*** [0.01]	0.02* [0.01]	0.01 [0.01]
$\Delta ASSETS$		0.20*** [0.03]		0.20*** [0.02]		0.10*** [0.03]		0.07** [0.03]
$\Delta SALES$		0.14*** [0.02]		0.10*** [0.02]		0.06** [0.03]		0.01 [0.02]
$\Delta CAPEX$		0.01* [0.01]		0.01 [0.01]		0.01 [0.01]		0.02* [0.01]
Other controls	yes	yes	yes	yes	yes	yes	yes	yes
Adj. R^2	0.03	0.05	0.03	0.05	0.05	0.07	0.04	0.06
No obs.	130,812	130,812	127,029	127,029	181,262	181,262	173,694	173,694

This table reports regressions of monthly small trade imbalances and monthly changes in short interest on changes in advertising spending and other control variables. The dependent variable in Columns 1–4 is the number of small buy orders minus the number of small sell orders, divided by the total number of small orders in a month, where small orders are defined as those below \$5,000 in size. In Columns 5–8, it is the monthly change in short interest (as a percentage of shares outstanding). The independent variable of interest is the change in log advertising spending (ΔAD). $\Delta ASSETS$, $\Delta SALES$, and $\Delta CAPEX$ are changes in log assets, sales, and capital expenditures in the same fiscal year. Other control variables include cumulative stock returns at various horizons, firm size, the market-to-book ratio, firm age, and average monthly turnover in the fiscal year. In Columns 1, 2, 5, and 6, the month small trade imbalance and change in short interest are measured in the same year as ΔAD , and in Columns 3, 4, 7, and 8, they are measured in the following year. Coefficients are estimated using the Fama-MacBeth approach. Standard errors, shown in brackets, are corrected for serial-dependence with twelve lags. *, **, and *** denote statistical significance at the 90%, 95%, and 99% level, respectively.

(b) Table 4: Retail order imbalance and short interest.

6.3 Table 5: Mechanism of the return effect and Table 6: Advertising spending around insider sales

Read: Table 5 shows stronger return reversals where the attention channel should be strongest: consumer firms, low analyst coverage, high retail buying, high turnover, low institutional ownership, and strong firm-product name overlap. Table 6 shows advertising rises before and during top-level insider sales and falls afterward, including in propensity-score matched comparisons.

Interpretation: Table 5 distinguishes attention from investment or growth-option interpretations. Table 6 then connects that attention effect to managerial incentives.

Economic message: Managers appear to exploit investors' limited attention when they personally benefit from temporary price pressure.

Table 5
Mechanism of the return effect

Monthly stock returns in year 1 (skip 6 months)							
(X 100)	[1]	[2]	[3]	[4]	[5]	[6]	[7]
ΔAD	-0.08*	-0.29**	-0.11*	-0.05	-0.17**	-0.23*	-0.11
	[0.05]	[0.12]	[0.07]	[0.09]	[0.07]	[0.12]	[0.12]
<i>Dummy</i>	0.16	-0.11	-0.21	-0.20	0.14	-0.19	0.25
	[0.17]	[0.15]	[0.20]	[0.14]	[0.17]	[0.23]	[0.25]
$\Delta AD * Dummy$ (Consumer Industry)	-0.18**						
	[0.08]						
$\Delta AD * Dummy$ (High Analyst Coverage)		0.21***					
		[0.07]					
$\Delta AD * Dummy$ (High Retail Buying)			-0.25**				
			[0.10]				
$\Delta AD * Dummy$ (High Turnover)				-0.21**			
				[0.08]			
$\Delta AD * Dummy$ (High Institution Holdings)					0.10*		
					[0.06]		
$\Delta AD * Dummy$ (High # Brands)						0.18**	
						[0.07]	
$\Delta AD * Dummy$ (Craigslist)							-0.21*
							[0.11]
Other controls	yes	yes	yes	yes	yes	yes	yes
Adj. R^2	0.09	0.11	0.12	0.10	0.09	0.06	0.09
No obs.	369,561	285,526	138,315	342,832	304,440	31,670	41,212

This table reports forecasting regressions of stock returns on changes in advertising spending and other control variables. The dependent variable is the monthly stock return in months 7–18 after the fiscal year-end. The independent variable of interest is the change in log advertising spending (ΔAD) in the previous fiscal year. $\Delta ASSETS$, $\Delta SALES$, and $\Delta CAPEX$ are changes in log assets, sales, and capital expenditures in the same fiscal year. Other control variables include cumulative stock returns at various horizons, firm size, the market-to-book ratio, firm age, and aggregate equity issuance in the previous four years as defined by [Daniel and Titman \(2006\)](#), discretionary accruals as defined by [Xid \(2001\)](#), and average monthly turnover in the fiscal year. The regressions also include a list of interaction terms between ΔAD and dummy variables. *Consumer industry* is an indicator variable that takes the value of one if the firm operates in a consumer-product industry based on the Fama-French five-industry definition and zero otherwise. In Columns 2–6, the indicator variable takes the value of one if the corresponding firm characteristic is above the sample median and zero otherwise. Retail trading is defined as the number of small buy orders minus that of small sell orders, divided by the total number of small orders in the previous year. *#brands* is the number of brands a firm has (as reported by Nielson Media). Analyst coverage, institutional ownership, and the number of product brands are all adjusted by firm size. Finally, *Craigslist* is a dummy that takes the value one if Craigslist is present in the same ZIP code as the firm's headquarters and zero otherwise. Coefficients are estimated using the Fama-MacBeth approach. Standard errors shown in brackets.

(a) Table 5: Mechanism of the return effect.

Table 6
Advertising spending around insider sales

Panel A: Panel regression of AD_t

	[1]	[2]	[3]
$PreEvent_t$	0.053***	0.053***	0.049***
	[0.016]	[0.018]	[0.018]
$Event_t$	0.069***	0.059***	0.059***
	[0.009]	[0.011]	[0.011]
$PostEvent_t$	-0.039***	-0.051***	-0.049***
	[0.015]	[0.018]	[0.018]
$SALES_{t-1}$	0.031***	0.198***	-0.658***
	[0.004]	[0.012]	[0.012]
$ASSETS_{t-1}$	0.006	0.496***	0.397***
	[0.005]	[0.015]	[0.015]
AD_{t-1}	0.940***		
	[0.003]		
Other controls	yes	yes	yes
Year fixed effects	yes	yes	yes
Firm fixed effects	no	yes	yes
Adj. R^2	0.95	0.94	0.80
No obs.	27,232	29,306	29,306

Panel B: Propensity score matching

	No obs.	Event firms - Matching firms
$AD_t - AD_{t-2}$	1,041	6.28%**
$AD_t - AD_{t-1}$	1,041	2.59%**
$AD_t - AD_{t+1}$	1,041	4.38%**
$AD_t - AD_{t+2}$	1,041	9.83%***

(Continued)

(b) Table 6: Insider sales, Panels A–B.

6.4 Table 6 continued and Table 7: Restricted-share vesting instrument

Read: Table 6 Panel C shows that advertising responses are larger when insider sale amounts are larger. Table 7 uses restricted-share vesting as an instrument for insider selling and finds that instrumented insider selling increases advertising.

Interpretation: These two tables address two important concerns. Panel C shows an intensity relation, and Table 7 uses a predetermined component of insider selling to reduce reverse-causality concerns.

Economic message: Advertising changes are not merely coincident with insider sales; they respond to incentives created by managers' impending equity liquidation.

Table 6
Continued

Panel C: Panel regression of AD_t

	[1]	[2]	[3]
$PreEvent_t$	0.020 [0.019]	0.029 [0.022]	0.029 [0.022]
$Event_t$	0.019* [0.010]	0.007 [0.013]	0.007 [0.013]
$PostEvent_t$	-0.028 [0.018]	-0.031 [0.022]	-0.028 [0.021]
$PreEvent_AMT_t$	0.873** [0.354]	0.524 [0.345]	0.524 [0.335]
$Event_AMT_t$	0.744*** [0.213]	0.911*** [0.306]	0.880*** [0.300]
$PostEvent_AMT_t$	-0.634** [0.270]	-1.245** [0.588]	-1.282** [0.579]
$SALES_{t-1}$	0.030*** [0.004]	0.198*** [0.012]	-0.658*** [0.012]
$ASSETS_{t-1}$	0.006 [0.005]	0.496*** [0.015]	0.397*** [0.015]
AD_{t-1}	0.939*** [0.003]		
Other Controls	yes	yes	yes
Year fixed effects	yes	yes	yes
Firm fixed effects	no	yes	yes
Adj. R^2	0.95	0.94	0.80
No obs.	27,232	29,306	29,306

This table reports analyses of advertising spending around insider sales. The sample period is 1986 to 2010. Panel A reports a panel regression. The dependent variable in Columns 1 and 2 is the logarithm of advertising spending (AD) in year t , and in Column 3, it is the logarithm of the advertising-spending-to-sales ratio. The independent variables of interest are three event-related dummies. A year is labeled an event year if the amount of insider sales (AMT) is above the 25th percentile of the sample distribution, where AMT is defined as the total number of shares sold by all top-level directors and offices in a year scaled by shares outstanding of the firm. $PreEvent$, $Event$, and $PostEvent$ are indicator variables, which are equal to one if the following year, the current year, and the previous year are an event year, respectively, and zero otherwise. $ASSETS$ and $SALES$ are the logarithm of total assets and sales, respectively. Other control variables include cumulative stock returns at various horizons, the return volatility, firm size, the market-to-book ratio, firm age, average monthly turnover in the fiscal year, and the KZ index. Year fixed effects are included in Column 1, and both year and firm fixed effects are included in Columns 2 and 3. Standard errors, shown in brackets, are clustered at the firm level. Panel B reports the differences in changes in log advertising spending around event years between the treatment group and matching firms. A matching firm is the one with the closest propensity score from the firm in question based on a set of firm characteristics: industry, firm size, book-to-market ratio, and past stock returns. Panel C reports a similar regression as in Panel A, with three additional independent variables: the actual amount of insider selling in the following ($PreEvent_AMT$), current ($Event_AMT$), and previous ($PostEvent_AMT$) years. *, **, and *** denote statistical significance at the 90%, 95%, and 99% level, respectively.

(a) Table 6 continued: insider-sale amount interactions.

Table 7
Vesting of restricted shares

2SLS regressions with restricted shares vesting

	[1]	[2]	[3]	[4]	[5]
	Baseline OLS		First stage	2SLS	
Dependent variable	AD_t	$(AD/SLSt)_t$	$Event_t$	AD_t	$(AD/SLSt)_t$
$Event_t$	0.048*** [0.018]	0.049*** [0.018]			
$Vesting_t$			0.598*** [0.160]		
$Instrumented\ Event_t$				0.058** [0.027]	0.062** [0.029]
$SALES_{t-1}$	0.134* [0.075]	-0.659*** [0.085]	0.038*** [0.006]	0.125* [0.071]	-0.705*** [0.077]
$ASSETS_{t-1}$	0.584*** [0.066]	0.457*** [0.068]	-0.013** [0.006]	0.580*** [0.064]	0.472*** [0.066]
Other controls	yes	yes	yes	yes	yes
Year fixed effects	yes	yes	yes	yes	yes
Firm fixed effects	yes	yes	yes	yes	yes
Adj. R^2	0.94	0.81	0.31	0.94	0.81
No obs.	8,544	8,544	8,544	8,544	8,544

This table reports advertising spending around insider sales using an instrument-variable approach. Columns 1 and 2 report the baseline OLS regression, where the dependent variables are the logarithm of advertising spending (AD_t) and the logarithm of the advertising-spending-to-sales ratio ($(AD/SLSt)_t$), respectively. Column 3 shows the first-stage regression, where the dependent variable is a dummy that takes the value of one if some insiders sell their shares in year t ($Event_t$). Columns 4 and 5 report the two-stage least squares regression, where the dependent variables are the logarithm of advertising spending (AD_t) and the logarithm of the advertising-spending-to-sales ratio ($(AD/SLSt)_t$), respectively. $Vesting_t$ is a dummy variable that takes the value of one if some restricted shares vest in year t . $Instrumented\ Event_t$ is the fitted value from the first-stage regression. $ASSETS$ and $SALES$ are the logarithm of total assets and sales, respectively. Other control variables include cumulative stock returns at various horizons, the return volatility, firm size, the market-to-book ratio, firm age, average monthly turnover, and the KZ index. Both year and firm fixed effects are included in all regression specifications. Standard errors, shown in brackets, are clustered at the firm level. *, **, and *** denote statistical significance at the 90%, 95%, and 99% level, respectively.

2, except that now the main independent variable is the instrumented insider sales from the first-stage regression. The coefficient on the instrumented event dummy of 5.8% is similar in magnitude to that reported in Columns 1 and 2. In sum, the evidence shown in Table 7 suggests a similar pattern in advertising

(b) Table 7: Restricted-share vesting.

6.5 Table 8: Sensitivity of future sales to advertising spending and Table 9: Advertising around sales by lower-level insiders

Read: Table 8 shows advertising predicts future sales growth in non-event years but the relation is significantly weaker or absent in insider-selling years. Table 9 shows no significant advertising pattern around equity sales by lower-level insiders.

Interpretation: Table 8 suggests insider-sale advertising is less productively motivated. Table 9 is a placebo-style test: lower-level insiders know operations but usually lack control over advertising budgets.

Economic message: Opportunistic advertising can be wasteful because it raises stock-market attention without producing comparable product-market sales growth.

Table 8
Sensitivity of future sales to advertising spending

Panel regression of $\Delta SALES_{t+1}$				
	[1]	[2]	[3]	[4]
ΔAD_t	0.085*** [0.005]	0.091*** [0.005]	0.027*** [0.005]	0.032*** [0.006]
$Event_t$		-0.063*** [0.006]		-0.069*** [0.006]
$\Delta AD_t \times Event_t$		-0.057*** [0.017]		-0.051*** [0.017]
$\Delta SALES_t$			0.115*** [0.015]	0.116*** [0.015]
$\Delta ASSETS_t$			0.262*** [0.011]	0.262*** [0.011]
Other controls	yes	yes	yes	yes
Year fixed effects	yes	yes	yes	yes
Firm fixed effects	no	no	no	no
Adj. R^2	0.07	0.08	0.20	0.21
No obs.	28,059	28,059	28,059	28,059

This table reports analyses of the sensitivity of future sales to advertising spending. The sample period is 1986 to 2010. The dependent variable in all columns is the percentage change in annual sales ($\Delta SALES$). The independent variable of interest is the lagged percentage change in advertising spending (ΔAD). A year is labeled an event year if the amount of insider sales (AMT) is above the 25th percentile of the sample distribution, where AMT is defined as the total number of shares sold by all top-level directors and offices in a year scaled by shares outstanding of the firm. $Event$ is an indicator variable, which is equal to one if the year in question is an event year, and zero otherwise. $\Delta ASSETS$ and $\Delta SALES$ are the percentage change in total assets and sales, respectively. Other control variables include cumulative stock returns at various horizons, the return volatility, firm size, the market-to-book ratio, firm age, average monthly turnover in the fiscal year, and the KZ index. Year fixed effects are included in all regression specifications. Standard errors, shown in brackets, are clustered at the firm level. *, **, and *** denote statistical significance at the 90%, 95%, and 99% level, respectively.

To test this idea, I conduct a simple regression analysis in which the dependent

(a) Table 8: Future-sales sensitivity.

Table 9
Advertising spending around sales by lower-level insiders

Advertising spending around sales by lower-level insiders			
	[1]	[2]	[3]
$PreEvent_t$	0.021 [0.015]	0.015 [0.019]	0.013 [0.019]
$Event_t$	0.019* [0.010]	0.018 [0.012]	0.020 [0.012]
$PostEvent_t$	-0.021 [0.018]	-0.019 [0.021]	-0.014 [0.021]
$SALES_{t-1}$	0.048*** [0.006]	0.176*** [0.028]	-0.677*** [0.026]
$ASSETS_{t-1}$	0.007 [0.004]	0.486*** [0.032]	0.386*** [0.031]
AD_{t-1}	0.923*** [0.004]		
Other controls	yes	yes	yes
Year fixed effects	yes	yes	yes
Firm fixed effects	no	yes	yes
Adj. R^2	0.94	0.94	0.79
No obs.	35,836	38,595	38,595

This table reports analyses of advertising spending around sales by lower-level insiders, as defined by Thomson Financial. The sample period is 1986 to 2010. The dependent variable in Columns 1 and 2 is the logarithm of advertising spending (AD) in year t , and in Column 3, it is the logarithm of the advertising-spending-to-sales ratio. The independent variables of interest are three event-related dummies. A year is labeled an event year if the amount of insider sales (AMT) is above the 25th percentile of the sample distribution, where AMT is defined as the total number of shares sold by all lower-level directors and offices in year t scaled by shares outstanding of the firm. $PreEvent$, $Event$, and $PostEvent$ are indicator variables, which are equal to one if the following year, the current year, and the previous year are an event year, respectively, and zero otherwise. $ASSETS$ and $SALES$ are the logarithm of total assets and sales, respectively. Other control variables include cumulative stock returns at various horizons, the return volatility, firm size, the market-to-book ratio, firm age, average monthly turnover in the fiscal year, and the KZ index. Year fixed effects are included in Column 1, and both year and firm fixed effects are included in Columns 2 and 3. Standard errors, shown in brackets, are clustered at the firm level. *, **, and *** denote statistical significance at the 90%, 95%, and 99% level, respectively.

product sales. I explore how corporate governance may mitigate this agency issue in Section 5.2.4.

(b) Table 9: Lower-level insiders.

6.6 Table 10: Other corporate events

Read: Table 10 Panel A shows advertising rises before and during equity issues but not debt issues. Panel B shows advertising rises around stock-financed acquisitions but not cash-financed acquisitions.

Interpretation: These results extend the insider-sale evidence to corporate transactions where firms benefit from a temporarily high stock price. The absence of similar patterns for debt issues and cash acquisitions is a useful placebo.

Economic message: Managers appear to use advertising when equity values are the transaction currency or when stock-sale proceeds matter.

suggest that the inverted v-shaped advertising pattern around equity sales is a robust phenomenon, unlikely to be driven solely by firms' investment opportunities and funding needs.

Table 10
Other corporate events

Panel A: Advertising spending around equity issues and debt issues

	[1]	[2]	[3]	[4]
<i>PreEvent_t</i>	0.038** [0.017]	0.019 [0.023]	0.012 [0.016]	0.004 [0.020]
<i>Event_t</i>	0.077*** [0.014]	0.041** [0.019]	0.025 [0.017]	0.016 [0.015]
<i>PostEvent_t</i>	0.008 [0.015]	0.021 [0.021]	0.023 [0.014]	0.027 [0.018]
<i>PreEvent_AMT_t</i>		0.205** [0.100]	0.032 [0.047]	0.033 [0.036]
<i>Event_AMT_t</i>		0.330*** [0.080]	0.033 [0.015]	0.033 [0.041]
<i>PostEvent_AMT_t</i>		-0.160** [0.083]	-0.015 [0.013]	0.296*** [0.015]
<i>SALES_{t-1}</i>	0.266*** [0.012]	0.266*** [0.012]	0.296*** [0.013]	0.296*** [0.013]
<i>ASSETS_{t-1}</i>	0.457*** [0.015]	0.457*** [0.015]	0.402*** [0.015]	0.403*** [0.015]
Other controls	yes	yes	yes	yes
Year fixed effects	yes	yes	yes	yes
Firm fixed effects	yes	yes	yes	yes
Adj. <i>R</i> ²	0.95	0.95	0.94	0.94
No obs.	26,111	26,111	23,381	23,381

(Continued)

Table 10
Continued

Panel B: Advertising spending around stock- and cash-financed mergers

	[1]	[2]	[3]	[4]
<i>PreEvent_t</i>	0.037 [0.050]	0.017 [0.062]	0.003 [0.019]	0.009 [0.025]
<i>Event_t</i>	0.140*** [0.042]	0.098* [0.052]	-0.017 [0.016]	-0.030 [0.020]
<i>PostEvent_t</i>	0.025 [0.044]	0.023 [0.056]	0.014 [0.020]	0.024 [0.026]
<i>PreEvent_AMT_t</i>		0.048 [0.051]		-0.047 [0.126]
<i>Event_AMT_t</i>		0.104** [0.053]		0.104 [0.092]
<i>PostEvent_AMT_t</i>		0.003 [0.042]		-0.069 [0.125]
<i>SALES_{t-1}</i>	0.208*** [0.030]	0.208*** [0.030]	0.274*** [0.017]	0.274*** [0.017]
<i>ASSETS_{t-1}</i>	0.414*** [0.040]	0.412*** [0.040]	0.447*** [0.021]	0.447*** [0.021]
Other controls	yes	yes	yes	yes
Year fixed effects	yes	yes	yes	yes
Firm fixed effects	yes	yes	yes	yes
Adj. <i>R</i> ²	0.94	0.94	0.96	0.96
No obs.	4,446	4,446	15,806	15,806

This table reports analyses of advertising spending around various corporate events: equity and debt issues in panel A, and stock- and cash-financed acquisitions in panel B (only acquirers are considered). The sample period of panel A is 1974 to 2010, and that in panel B is 1980 to 2010. The dependent variable in all columns of both panels is the logarithm of advertising spending (*AD*) in year *t*. The independent variables of interest are three event-related dummies. A year is labeled an event year if there is at least one transaction of the particular type under consideration. In panel A, *AMT* is defined as the total proceeds from equity issues (Columns 1 and 2) and debt issues (Columns 3 and 4) in year *t* scaled by the market capitalization of the firm. In panel B, *AMT* is defined as the total transaction value of stock-financed acquisitions (Columns 1 and 2) and cash-financed acquisitions (Columns 3 and 4) in year *t* scaled by the market capitalization of the firm. *PreEvent*, *Event*, and *PostEvent* are indicator variables, which are equal to one if the following year, the current year, and the previous year are an event year, respectively, and zero otherwise. *PreEvent_AMT*, *Event_AMT*, and *PostEvent_AMT* are the actual *AMT* in the following, current, and previous years, respectively. *ASSETS* and *SALES* are the logarithm of total assets and sales, respectively. Other control variables include cumulative stock returns at various horizons, the return volatility, firm size, the market-to-book ratio, firm age, average monthly turnover in the fiscal year, and the KZ index. Both year and firm fixed effects are included in all columns. Standard errors, shown in brackets, are clustered at the firm level. *, **, and *** denote statistical significance at the 90%, 95%, and 99% level, respectively.

(a) Table 10 Panel A: Equity and debt issues.

(b) Table 10 Panel B: Stock- and cash-financed mergers.

7 Interpretive synthesis

- **Step 1: advertising attracts attention.** Increased advertising spending is associated with retail buying and contemporaneous return increases.
- **Step 2: prices overshoot.** The following-year and two-year returns are negative, consistent with temporary overvaluation rather than permanent cash-flow news.
- **Step 3: managers understand the channel.** Advertising rises before and during top-level insider sales and other equity-sensitive events.
- **Step 4: opportunistic advertising is less productive.** The sales-growth link weakens in insider-sale years.
- **Step 5: placebo patterns fit.** Lower-level insiders, debt issues, and cash acquisitions do not show the same strong pattern.

8 Short summary

- The paper studies whether product advertising affects investor attention and stock returns.
- Firms with high advertising-growth experience high contemporaneous returns and negative future returns.
- Retail investors buy high-advertising-growth stocks, and short interest also rises.
- The return effect is stronger when advertising is more likely to reach attention-constrained investors.
- Advertising rises before and during top-level insider sales and falls afterward.
- Restricted-share vesting supports the opportunistic-advertising interpretation.
- Advertising around insider sales has weaker future-sales effects.

- Similar patterns occur for seasoned equity offerings and stock-financed acquisitions, but not for debt or cash acquisitions.

9 Conclusion

9.1 Core conclusion

Lou shows that product-market advertising can spill over into financial markets by attracting investor attention. The resulting attention creates temporary price pressure and return reversal.

9.2 Incremental contribution in one sentence

The paper's distinctive contribution is to connect a real investment decision—advertising—to investor attention, temporary mispricing, and managerial opportunism around equity sales in a single empirical framework.

9.3 Economic intuition

Advertising makes the firm more salient to retail investors. Attention-constrained retail investors buy the stock, causing temporary price pressure. Managers who recognize this effect have incentives to increase advertising before selling equity or using stock as a transaction currency.

9.4 Takeaway

Advertising can be a financial-market instrument as well as a product-market instrument. The paper therefore links behavioral asset pricing, corporate investment, and managerial agency problems.